

CONTACT
INFORMATION

Adrian William Romero Jorge

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EDUCATION

- Ph.D. Student**, Goethe University Frankfurt, Germany.
Supervisor: Prof. Dr. Elena Bratkovskaya, Prof. Dr. Laura Sagunski **2023 - present**
- M.S. Physics**, University of Havana, Cuba.
Supervisors: Prof. Dr. A. Martínez Pérez, Prof. Dr. E. Rodríguez Querts **2022 - 2023**
- B.S. Physics**, University of Havana, Cuba.
Supervisors: Prof. Dr. A. Martínez Pérez, Prof. Dr. E. Rodríguez Querts. Five-years course plus a thesis **2016 - 2021**

POSTGRADUATES
COURSES

- University of Havana & Higher Institute of Technologies and Applied Sciences, Cuba.**
- Introduction to General Relativity Theory
 - Automation of Physics Experiments with the Arduino Platform
 - Standard Model of Particles. Mathematical Methods
 - Introduction to Quantum Fields in Classic Contexts
 - Quantum Field Theory
 - Cosmology
 - Introduction to Standard Model of Particles
 - Nonlinear Quantum Electrodynamics

COMPLEMENTARY
EDUCATION

- Institute for Theoretical Physics, Goethe University, Frankfurt, Germany.**
- Physics of Strongly Interacting Matter
 - Dynamical models for relativistic heavy-ion collisions

Galileo Galilei Institute (GGI), Italy

- School: Direct Detection across Dark Matter mass ranges 2025 (TME)
- School: Frontiers in Nuclear and Hadronic Physics 2026 (FNHP)

Helmholtz Graduate School for Hadron and Ion Research (HGS-HIRE), Germany.

- Making an Impact as an Effective Researcher
- Career and Leadership
- Leading Teams in a Research Environment
- Machine Learning from Scratch

PRESENTATIONS
& EVENTS

- Theory Meets Experiments 2025: Dark Matter Direct Detection (TME)**
Lightning Talk: "Search for dark photons in heavy-ion collisions". Florence, Italy. **2025**
- IV DarkMatter@Bonn Meeting (DMLab)**
Talk: "Combined constraints on dark photons from high-energy collisions, cosmology, and astrophysics". Bonn, Germany. **2025**
- V Dark Matter Conference: From the Smallest to the Largest Scale (DM)**
Poster and Flash Talk: Constraining Dark Photon Kinetic Mixing in Heavy-Ion Collisions from

SIS to LHC energies. Santander, Spain. 2025

XXXI International Conference on Ultra-relativistic Nucleus-Nucleus Collision (QM)

Poster: Electromagnetic emission from strongly interacting hadronic and partonic matter created in heavy-ion collisions. Frankfurt, Hesse, Germany 2025

III Workshop on Heavy Ions Physics and Compact Stars (HIPSTARS)

Talk: "Exploring Dark Photon Production and Kinetic Mixing Constraints in Heavy-Ion Collisions ". ICIMAF, Havana, Cuba. 2024

42nd International Conference on High Energy Physics (ICHEP)

Talk: "Search for dark photon in heavy ion collisions ". Prague, Czech Republic. 2024

GET INvolved Program by GSI-FAIR

Research training with Dr. Elena Brakovskaya. GSI, Darmstadt, Germany. 2022

II Workshop on Heavy Ions Physics and Compact Stars (HIPSTARS).

Talk: "On the propagation of photons in rotating neutron stars including quantum and gravitational effects". ICIMAF, Havana, Cuba. 2022

Proposals for the Award of the Nuclear Energy Agency and Advanced Technologies (AENTA).

Talk: "Delay of the magnetized photon in astrophysical scenarios". Cuban National Hotel, Havana, Cuba. 2021

I Workshop on Heavy Ions Physics and Compact Stars (HIPSTARS).

Talk: "The photon time delay in strong magnetic field". ICIMAF, Havana, Cuba. 2020

XV Symposium of the Cuban Physics Society.

Poster: "The photon time delay in magnetized vacuum magnetosphere". University of Havana, Havana, Cuba. 2020

VI International Sym. on Strong Electromagnetic Fields and Neutron Stars SMFNS). and V Caribbean Sym. on Cosmology, Gravitation, Nuclear and Astroparticle Physics (STARS).

Talk: "The magnetized photon time delay". Varadero, Cuba. 2019

SCHOOLS

III German-Cuban Summer School by DAAD-FAIR Program.

Participation. ICIMAF, Havana, Cuba. 2023

European Summer School on Science Communication (ES3C) by DAAD-FAIR Program.

Participation. FIAS, Frankfurt, Germany. 2022

II German-Cuban Summer School by DAAD-FAIR Program.

Talk: "Photon time delay in neutron stars magnetosphere". ICIMAF, Havana, Cuba. 2022

I German-Cuban Summer School by DAAD-FAIR Program.

Participation. Havana, Cuba. 2019

WORKSHOPS AS
UNDERGRADUATE
STUDENT

- Student Scientific Forum of Havana University.**
Participation. University of Havana, Havana, Cuba. **2022**
- XXXVII Student Scientific Workshop – Physics Faculty.**
*Talk: “Photon time delay in neutron stars magnetosphere”.
Faculty of Physics, University of Havana, Havana, Cuba.* **2021**
- VI Student Congress of Scientific and Technological Research (CEICYT).**
*Online Talk: “Photon time delay modeling in the magnetosphere of pulsars with
ultra-intense magnetic fields”. Dominican Republic.* **2021**
- XXXVI Student Scientific Workshop – Physics Faculty.**
Participation. Faculty of Physics, University of Havana, Havana, Cuba. **2020**
- XXXV Student Scientific Workshop – Physics Faculty.**
*Talk: “Delay of magnetized photons”.
Faculty of Physics, University of Havana, Havana, Cuba.* **2019**
- XXXIV Student Scientific Workshop – ICIMAF.**
*Talk: “Delay of photons in magnetized vacuum”.
Faculty of Physics, University of Havana, Havana, Cuba.* **2018**
- XXXIII Student Scientific Workshop – Physics Faculty.**
*Talk: “Quantum dynamics of vibrations between the rings of benzene”.
Faculty of Physics, University of Havana, Havana, Cuba.* **2017**

PEER-REVIEWED
PUBLICATIONS

- A. W. Romero Jorge**, L. Sagunski, Guan-Wen Yuan, T. Song and E. Brakovskaya. “*Combined constraints on dark photons from high-energy collisions, cosmology, and astrophysics*”, Phys. Rev. D 113, **113**, 055052, arXiv.2601.15066, 2026.
- A. W. Romero Jorge**, E. Brakovskaya, T. Song and L. Sagunski, “*Constraints on Kinetic Mixing of Dark Photons from Dilepton Spectra*”, Phys. Rev. C, **112**, 054905, arXiv.2507.11163, 2025.
- A. W. Romero Jorge**, T. Song, Q. Zhou and E. Brakovskaya, “*Electromagnetic emission from strongly interacting hadronic and partonic matter created in heavy-ion collisions*”, Phys. Rev. C, **111**, 064904, arXiv.2503.05253, 2025.
- A. W. Romero Jorge**, A. Pérez Martínez, and E. Rodríguez Querts, “*Nonlinear electrodynamics for the vacuum of Dirac materials. Photon magnetic properties and radiation pressures*”, Phys. Rev. A, **110**, 012201, arXiv.2402.01042, 2024.

CONFERENCE
PROCEEDINGS

- A. W. Romero Jorge**, E. Brakovskaya, T. Song, and L. Sagunski, “*Exploring Dark Photon Production and Kinetic Mixing Constraints in Heavy-Ion Collisions*”, Astron. Nachr., **346**: e20240132, arXiv.2412.02536, 2024.
- A. W. Romero Jorge**, E. Brakovskaya, and L. Sagunski, “*Search for dark photons in heavy-ion collisions*”, Proceedings of 42nd International Conference on High Energy Physics, PoS (ICHEP2024), **290**, arXiv.2409.20141, 2024.
- A. Pérez Martínez, M. Pérez García, E. Rodríguez Querts and **A. W. Romero Jorge**, “*Vacuum properties and astrophysical implications*”, 16th Marcel Grossmann Meeting, 3756-3761, 2023.
- A. W. Romero Jorge**, E. Rodríguez Querts, A. Pérez Martínez and H. Pérez Rojas, “*The photon*

time delay in magnetized medium magnetosphere", Astron. Nachr., **342**, 382-387, 2021.

A. W. Romero Jorge, E. Rodríguez Querts, A. Pérez Martínez et al., "*The photon time delay in magnetized vacuum magnetosphere*", Astron. Nachr., **340**, 852-856, arXiv.1912.02904 2020.

HONORS
& AWARDS

Giersch Excellence Award 2025.

The grant is awarded for outstanding work in the thesis project in the past year. **2024, 2025**

Karin and Carlo Giersch Scholarship.

Awarded for Ph.D. studies in physics **2023**

Graduated with Honors (summa cum laude).

Awards for the best graduate of the career and most outstanding results in the scientific research. Faculty of Physics, University of Havana. **2021**

Award for the scientific result of the science and technological innovation system of the Nuclear Energy and Advance Technologies Agency (AENTA). **2021**

*To the work "Magnetized photon delay in astrophysical scenarios" by the authors **A. W. Romero Jorge**, E. Rodríguez Querts and A. Pérez Martínez.*

First place in the Student Scientific Workshop – Physics Faculty.

At the theoretical physics committee. **2021**

TEACHING
EXPERIENCE

Faculty of Physics, University of Havana, Havana, Cuba.

Teaching Assistant **2018-2019**

Experimental physics laboratory (Mechanics) and on the computational methods course (C++).

High School Institute José Martí, Havana, Cuba.

Part time teacher **2019**

Teaching Classical Mechanics to 10th grade.

PROFICIENCY

Languages	:	Spanish (native), English (oral,writing: fluent), German (A1), Japanese (begginer)
Programming Languages	:	Fortran, C++, Python, Java, XML, Arduino, Wolfram
Softwares/Tools	:	L ^A T _E X, Origin, Wolfram Mathematica, Maple, Android Studio,
Hard Skills	:	Drawing, Video Editing, Digital Design, Web & App development
Soft Skills	:	Teamwork, Adaptability, Time management, Creativity, Decision-making, Analytical thinking, Attention to detail